

Spider Bot using Theo Jansen Mechanism 20ME6DCMW2

Research Project Work (20ME5DCMW2) report submitted to



Visvesvaraya Technological University,

Belagavi in partial fulfilment of the

requirements for the degree of

Bachelor of Engineering in Mechanical Engineering

by

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Certificate

Certified that the research project work entitled "Spider Bot using Theo Jansen Mechanism" is a bonafide record of work done carried out by

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in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Mechanical Engineering of the Visvesvaraya Technological University, Belagavi, during the year 2022 - 23. It is certified that all corrections/suggestions indicated during the internal assessments have been incorporated.

Signature of Guide
(Dr. G. Giridhara)

Signature of HOD
(Dr. G. Giridhara)

Signature of Principal
(Dr. S. Muralidhara)

Semester End Examination

Name of Examiner

Signature with Date

- 1.
- 2.

Declaration

We, hereby declare that the **Research** project work entitled "Spider Bot using Theo Jansen Mechanism" has been carried out by us under the guidance of Dr. G. Giridhara, Professor, Department of Mechanical Engineering, B. M. S. College of Engineering, Bengaluru, and in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Mechanical Engineering of Visvesvaraya Technological University, Belagavi.

We further declare that we have not submitted this report either in part or in full to any other university for the award of any degree/diploma.

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Place: Bengaluru

Date:28/07/2023

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Abstract

Legged robots that imitate the of living creatures have the ability to travel or climb over terrain which is inaccessible for a wheeled or followed vehicle. Autonomous legged robots have a great future because they are frequently used for space missions on other worlds and in hazardous locations, like inside an atomic pile. Low power consumption and weight are further advantages of walking robots; therefore, it is important to use the minimum number of actuators. The Theo Jansen mechanism is gaining widespread popularity among legged robotics researchers due to its scalable design, energy efficiency, low payload-to-machine load ratio, bio-inspired locomotion, deterministic foot trajectory and other advantages.

In this project, detailed design calculations were performed for the Theo-Jansen leg mechanism, joint expeller, bending moments and associated structural elements. The leg and body components of the spider bot were modeled in SolidWorks, and static as well as dynamic simulations were carried out to evaluate stresses, deformations and motion characteristics on varied terrains. Kinematic analysis of the leg mechanism was conducted in MATLAB to obtain the locus of key points of the Theo-Jansen leg and to verify the desired walking trajectory. The physical model was fabricated using laser cutting for the frame and links, and 3D printing for critical joints and connectors. An Arduino-based embedded system was then implemented, integrating an Arduino board, L298N motor driver, DC motors and batteries, with custom Arduino code for forward motion, turning and navigation. The performance of the spider bot using the Theo Jansen mechanism was experimentally evaluated on hard, rough and muddy terrains to study its locomotion behavior and surface interaction.

Chapter 1

Introduction

1.1 Introduction to Kinematics

Kinematics is a subfield of physics developed in classical mechanics that describes the motion of points, bodies (objects), and systems of bodies (groups of objects) without considering the forces that cause them to move. Kinematics, as a field of study, is often referred to as the "geometry of motion" and is occasionally seen as a branch of mathematics. Kinematics is the study of how things move, while dynamics is the study of forces and motion. Kinematics deals with the relationship between an object's position and its velocity or speed. Dynamics refers to all physical phenomena that involve a change in momentum, such as movement, acceleration, etc. The Fundamental theorem of kinematics states that if a body moves along a straight line with constant speed, then its acceleration is zero. A kinematics problem begins by describing the geometry of the system and declaring the initial conditions of any known values of position, velocity and/or acceleration of points within the system. Then, using arguments from geometry, the position, velocity and acceleration of any unknown parts of the system can be determined. The study of how forces act on bodies falls within kinetics, not kinematics. In mechanical engineering, robotics, and biomechanics kinematics is used to describe the motion of systems composed of joined parts (multi-link systems) such as an engine, a robotic arm or the human skeleton.

Geometric transformations, also called rigid transformations, are used to describe the movement of components in a mechanical system, simplifying the derivation of the equations of motion. They are also central to dynamic analysis. Kinematic analysis is the process of measuring the kinematic quantities used to describe motion. In engineering, for instance, kinematic analysis may be used to find the range of movement for a given mechanism and, working in reverse, using kinematic synthesis to design a mechanism for a desired range of motion. In addition, kinematics applies algebraic geometry to the study of the mechanical advantage of a mechanical system.

1.2 SpiderBot

Spider Bots are a type of legged robots are a type of mobile robot which use articulated limbs, such as leg mechanisms, to provide locomotion. They are more versatile than wheeled robots and can traverse many different terrains, though these advantages require increased complexity and power consumption. Legged robots often imitate legged animals, such as humans or insects, in an example of biomimicry. Legged robots, or walking machines, are designed for locomotion on rough terrain and require control of leg actuators to maintain balance, sensors to determine foot placement and planning algorithms to determine the direction and speed of movement. The periodic contact of the legs of the robot with the ground is called the gait of the walker.

In order to maintain locomotion the center of gravity of the walker must be supported either statically or dynamically. Static support is provided by ensuring the center of gravity is within the support pattern formed by legs in contact with the ground. Dynamic support is provided by keeping the trajectory of the center of gravity located so that it can be repositioned by forces from one or more of its legs. Legged robots can be categorized by the number of limbs they use, which determines gaits available. Many-legged robots tend to be more stable, while fewer legs lends itself to greater maneuverability.

1.3 Leg Mechanism

Leg Mechanism: A leg mechanism (walking mechanism) is a mechanical system designed to provide a propulsive force by intermittent frictional contact with the ground. A robot leg is a mechanical leg that performs the same functions that a human leg can. The robotic leg is typically programmed to execute similar functions as a human leg. A robotic leg is similar to a prosthetic leg. However, a robotic leg can be controlled electrically or mechanically. This is in contrast with wheels or continuous tracks which are intended to maintain continuous frictional contact with the ground. Mechanical legs are linkages that can have one or more actuators and can perform simple planar or complex motion. Compared to a wheel, a leg mechanism is potentially better fitted to uneven terrain, as it can step over obstacles. . Legged robots use articulated limbs, such as leg mechanisms, to provide locomotion.

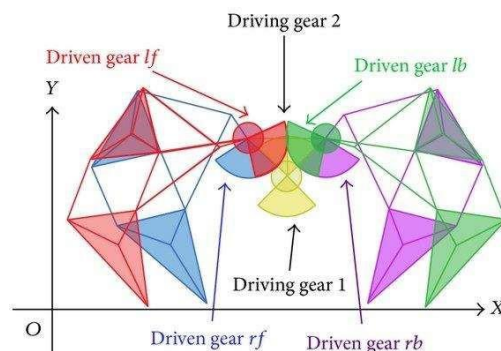


Figure 1. Theo Jansen Mechanism schematic

1.4. Types of leg Mechanisms

1. Jansen Mechanism

Theo-Jansen mechanism-Jansen linkage is a planar leg mechanism designed by the kinetic sculptor Theo Jansen to simulate a smooth walking motion . Jansen has used his mechanism in a variety of kinetic sculptures which are known as Jansen's linkage bears artistic as well as mechanical merit for its simulation of organic walking motion using a simple rotary input These leg mechanisms have applications in mobile robotics and in gait analysis.

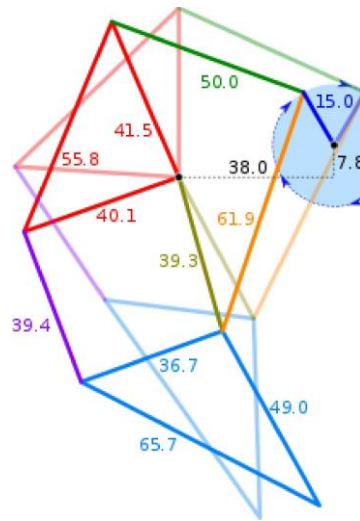


Figure 2.Theo Jansen Leg

2.Klann Mechanism

Klann mechanism-The Klann linkage is a planar mechanism designed to simulate the gait of legged animal and function as a wheel replacement, a leg mechanism. The linkage consists of the frame, a crank, two grounded rockers, and two couplers all connected by pivot joints. It was developed by Joe Klann in 1994 as an expansion of Burmester curves which are used to develop four-bar double rocker linkages such as harbor crane booms.

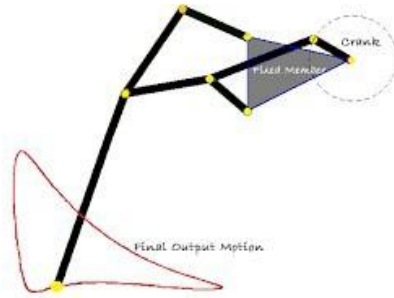


Figure 3. Klann mechanism Leg

3.Strider Mechanism

strider mechanism-A strider mechanism (walking mechanism) is a mechanical system designed to provide a propulsive force by intermittent frictional contact with the ground. This is in contrast with wheels or continuous tracks which are intended to maintain continuous frictional contact with the ground.

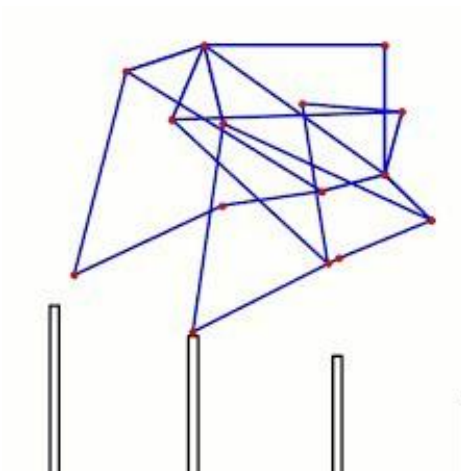


Figure 4. Strider mechanism leg

4.Plantigrade Mechanism

Plantigrade mechanism-In kinematics, the Chebyshev Lambda Linkage is a four-bar linkage that converts rotational motion to approximate straight-line motion with approximate constant velocity. It is so-named because it looks like a lowercase Greek letter lambda(λ). The precise design trades off straightness, lack of acceleration, and the proportion of the driving rotation that is spent in the linear portion of the full curve.

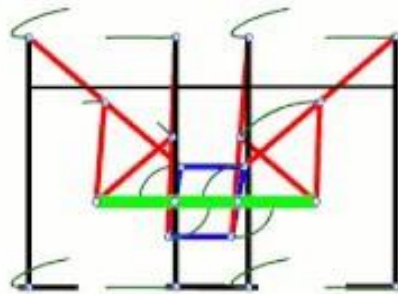


Figure 5. Plantigrade Mechanism Leg

1.5 Advantages of Theo Jansen Mechanism

- 1.horizontal speed as constant as possible while touching the ground (support phase)
- 2.while the foot is not touching the ground, it should move as fast as possible
- 3.constant torque/force input (or at least no extreme spikes/changes)
- 4.stride height (enough for clearance, not too much to conserve energy)
- 5.the foot has to touch the ground for at least half of the cycle for a two/four leg mechanism or respectively, a third of the cycle for a three/six leg mechanism
- 6.minimized moving mass
- 7.vertical center of mass always inside the base of support
- 8.the speed of each leg or group of legs should be separately controllable for steering
- 9.the leg mechanism should allow forward and backward walking.

Chapter 2

Literature Survey

The walking spider bot is a fascinating field of robotics that aims to replicate the locomotion and behavior of spiders in an artificial system. It involves designing and developing a multi-legged robot capable of walking, climbing, and navigating through challenging terrains. To understand the current state of the art and gather insights into the existing research and developments, a comprehensive literature survey was conducted. The survey covered relevant academic papers, research articles, conference proceedings, and books published up until the knowledge cutoff date of September 2021.

The Literature surveyed by us for increasing our understanding and help us gain insight into the subject consists of mainly four papers surveyed by us.

The first paper titled "**Kinematic Analysis of Theo Jansen Mechanism**" authored by Austin Pavlak investigates the kinematic behaviour and the characteristics of the Theo Jansen Mechanism.

The second paper titled "**Dynamic Analysis of Theo Jansen Mechanism**" authored by Jenny Zhou speaks about the features and characteristics of the Theo Jansen Mechanism when subjected to dynamic analysis using an analysis software.

The third paper used to identify advantages of the mechanism was "**Analysis and advantages of Theo Jansen**" By Mohan Rajesh speaks about the advantages of the Theo Jansen Mechanism when compared to other types of mechanisms.

The Last paper referred titled "**Material selection Analysis for legged Robots**" By Swadeen Patnaik is related to the analysis and selection procedure for materials used in legged mechanisms. Therefore, the Literature survey presented a comprehensive overview in the fields of modelling, analysis , material selection and testing.

Chapter 3

Aim and Objective

3.1 Aim of the Project

The aim of this project is to design, analyze, fabricate and control a spider bot based on the Theo Jansen leg mechanism, and to evaluate its locomotion behavior and stability on different types of terrain using combined analytical, simulation and experimental approaches.

3.2 Objectives

1. Design and Modeling

- To perform design calculations for the Theo Jansen leg mechanism, including link dimensions, joint forces and bending moments in critical members.
- To model the leg components and body structure of the spider bot using SolidWorks and generate complete 3D assemblies and detailed drawings.

2. Kinematic and Structural Analysis

- To carry out kinematic analysis of the Theo Jansen leg using MATLAB in order to obtain and study the locus of the leg tip over a full cycle.
- To perform static and motion (dynamic) analyses in SolidWorks to evaluate stresses, deformations and motion characteristics of the spider bot under typical loading conditions.

3. Fabrication

- To fabricate the spider bot by laser cutting the frame and leg links, and by 3D printing joints and other critical components, ensuring accurate realization of the designed geometry.

4. Control and Navigation

- To implement an Arduino-based control system integrating an Arduino board, L298N motor driver, DC motors and batteries, and to develop Arduino code for forward motion, reverse motion and turning of the spider bot using differential drive.

5. Experimental Evaluation on Different Terrains

- To test the fabricated spider bot on hard, rough and muddy terrains, and to compare the experimental performance (stability, speed, slippage and adaptability) with the predictions from MATLAB kinematic analysis and SolidWorks simulations.

Chapter 4

Methodology

The Methodology followed by us after consists of the following steps which are-

- (i) Modelling
- (ii) Material Selection
- (iii) Analysis
- (iv) Fabrication
- (v) Testing

4.1-Dimensional Modeling

Front view

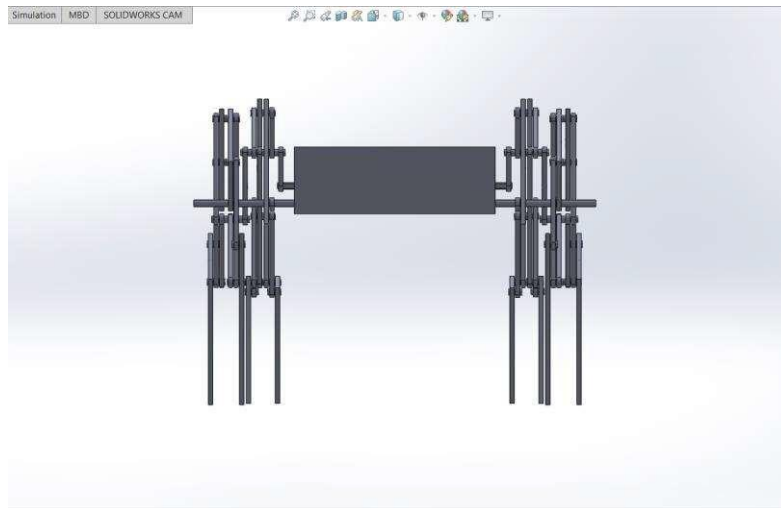


Figure 6: Front View

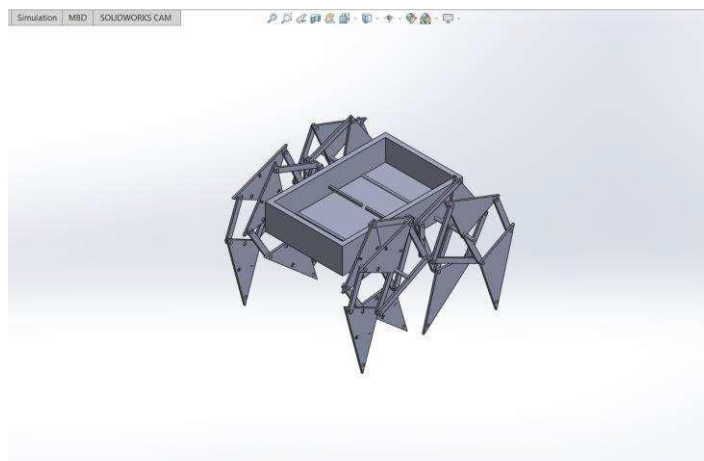


Figure 7: Isometric View

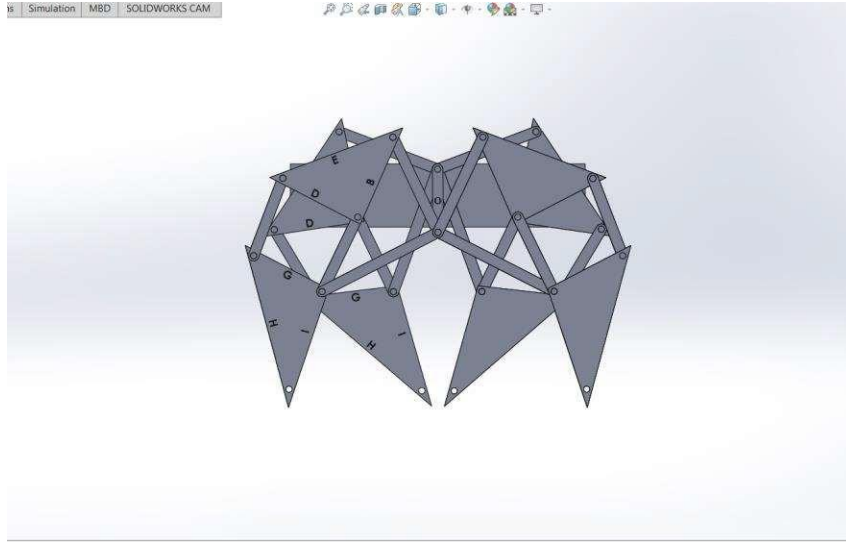


Figure 8 :Side View

4.2. Material Selection

Material selection is the process of choosing the best materials for a particular product or application based on various factors such as mechanical properties, durability, availability, cost, and environmental impact. Material selection is critical in engineering design and plays a crucial role in the overall performance, quality, and functionality of a product. The selection of materials is influenced by several factors such as the application, manufacturing process, and service conditions, and requires a thorough understanding of material properties and behavior. The overall project itself is to develop an integrated surface vehicle system, or robot. The robot's purpose is to be used for bomb disposal in military applications. This robot must meet certain requirements, including: a capture carry system that is able to pickup three objects of various sizes, the ability to detect and capture metal, the ability to move in various terrains, the ability to move over objects such as a log, stream live video, and a few other components. In order to fulfill as many of these requirements as possible, the robot frame design must be able to house all of the electrical circuits, motors, batteries, and other elements necessary for an operational robot. Thus, the frame must be made of a strong, rigid material that will not break under additional weight and can withhold additional stress at many different points on the frame. In working on this project, one of my subsystems involved the frame material selection and frame design that was used as the structural basis for the robot.

4.2.1. Material Selection Process

The first step involves translating the design requirements. This step involves taking the design requirements and translating the criterion into a material. The first step in translation is deriving the function of the design. In this case, the function is the robot frame that must be sturdy enough to remain in one piece while housing all of the other components for the competition. The function of the material must then meet the given constraints of the design, such as the weight, dimensions, and other components. The objective of the design process can be determined. The objective involves what you are trying to maximize or minimize with the material choice, such as maximizing the strength of the material. In order to reach the objective of the design, there are normally certain variables that are open to interpretation and can change in order to better fit the objective. These variables are called free variables and could include allowing the thickness of a plate to vary in order to optimize the objective. The translation phase is crucial in determining what materials to select because it allows the designer to lay out all of the design requirements in order to optimize the material selection. The second step in the materials selection process involves screening. In the screening phase, materials are eliminated that absolutely will not fit the engineering problem. The screening is done by looking at the constraints found in the translation phase. If a material does not meet the requirements of the design constraints, then the material is not suitable for this application. An example could be in that the constraints for a rod must meet a certain strength requirement. All of the materials that don't match this strength requirement are then eliminated from discussion because it will not fit the functionality of the design. The next step in the material selection process involves ranking.

Ranking looks at the materials that were approved during screening and see which ones best fit the objective of the engineering design. In order to do this, a material index must be derived in order to optimize the performance of the material. The material index is the best way to rank the materials that passed the screening step. The material index can reflect the performance of a single property or multiple properties depending on the design requirements and the objectives. The final step in the material selection process involves finding the proper supporting information.

4.2.2. Material Selected - FibreBoard

Fibreboard, also known as fiberboard or particle board, is a type of engineered wood product that is widely used in various industries and applications. It is made from wood fibers, such as sawdust or wood chips, which are bonded together using resin or other synthetic materials. Here is a brief overview of fibreboard:

Composition: Fibreboard is composed of wood fibers that are mechanically processed and compressed under high pressure and temperature. The fibers are mixed with a binder, typically a synthetic resin, to provide cohesion and strength to the board.

Types of Fibreboard: There are different types of fibreboard available, including:

1. **Medium Density Fibreboard (MDF):** MDF has a dense and uniform composition, making it suitable for a wide range of applications. It is commonly used in furniture manufacturing, cabinetry, and interior decor.
2. **High Density Fibreboard (HDF):** HDF is similar to MDF but has a higher density and greater strength. It is often used in applications where durability and resistance to impact are important, such as flooring and doors.
3. **Low Density Fibreboard (LDF):** LDF is less dense than MDF and is commonly used in applications where weight is a concern, such as in packaging materials.

Advantages of Fibreboard -

1. **Affordability:** Fibreboard is relatively inexpensive compared to solid wood or other engineered wood products, making it a cost-effective option for many applications.
2. **Versatility:** It can be easily shaped, cut, and machined into various forms and sizes, allowing for flexibility in design and manufacturing.
3. **Consistency:** Fibreboard offers uniform density and strength throughout the board, ensuring predictable performance and easy handling.
4. **Eco-Friendly:** It utilizes wood waste and by-products, reducing the demand for virgin timber and promoting sustainable forestry practices.

4.2.3. Properties of Fibreboard

1.Density: Fibreboard is available in different densities, ranging from low-density to high-density variants. The density affects the strength, weight, and other performance characteristics of the board. Higher density fibreboard generally offers better strength and durability.

2.Strength and Stability: Although fibreboard is not as strong as solid wood, it still exhibits reasonable strength and stability. It has good resistance to bending and can hold its shape well under normal conditions. High-density fibreboard tends to be stronger and more rigid than low-density variants.

3.Machinability: Fibreboard is relatively easy to work with using common woodworking tools. It can be cut, drilled, shaped, and routed to achieve desired dimensions and designs. The consistency of the material allows for precision machining and clean edges.

4.Surface Finish: Fibreboard has a smooth and uniform surface, which makes it suitable for applications where a polished or painted finish is desired. It accepts paint, veneers, laminates, and other surface treatments well, providing a versatile range of decorative options.

5.Sound and Thermal Insulation: Due to its composition and density, fibreboard exhibits good sound absorption and thermal insulation properties. It can help reduce noise transmission and improve energy efficiency in buildings.

6.Dimensional Stability: Fibreboard is less prone to shrinking, warping, or expanding due to changes in temperature and humidity compared to solid wood. This dimensional stability makes it a suitable choice for applications where consistent dimensions are crucial.

7.Fire Resistance: Some fibreboard products may have inherent fire-resistant properties, which can contribute to fire safety in certain applications. However, it is essential to check specific fire ratings and certifications for the intended use.

4.3 Analysis

4.3.1 Kinematic Analysis

Kinematic analysis of a mechanism consists of calculating position, velocity and acceleration of any of its points or links.

To numerically evaluate the motion of the Theo Jansen leg and to obtain the locus of the leg tip for one complete crank rotation, a kinematic simulation was implemented in MATLAB. The script uses the specified link lengths and joint constraints to compute the positions of key points of the mechanism as a function of crank angle. By sweeping the crank through 0–360°, the Cartesian coordinates of the leg tip are calculated and plotted, allowing clear visualization of the support and swing phases of the gait. The MATLAB code used for this analysis is provided below (and in Appendix X for reference).

```
clear ; clc; close all;

L1=5.5;L2=1;L3=7.5;L33=7.5;L4=4.2;L44=4.2;L5=4.0;L6=4.2;L66=7.5;L7=4.2;
A1=0;A2=pi/2:0.3:6*pi+(pi/2);

A2=-A2;

%% *****POSITION ANALYSIS*****

% LOCATION OF POINT B
Bx=L2.*cos(A2);
By=L2.*sin(A2);

% FIRST CASE OF 4 BARS
[A3,A4]=Caso_4b(L1,L2,L3,L4,A1,A2,'cruzado');
A5=(pi-A4)+(pi/2);

% LOCATION OF POINT C
Cx=Bx+(L3.*cos(A3));
Cy=L4.*sin(A4);

% LOCATION OF POINT D
Dx=L1; Dy=0;

% SECOND CASE OF 4 BARS
p=2.*L4.*((L1.*cos(A1))-(L2.*cos(A2)));
q=2.*L4.*((L1.*sin(A1))-(L2.*sin(A2)));
r=(L1.^2)+(L2.^2)-(L33.^2)+(L44.^2)-(2.*L1.*L2.*cos(A1-A2));

A44=2.*atan((-q+sqrt((p.^2)+(q.^2)-(r.^2)))/(r-p));

d=2.*L3.*((L2.*cos(A2))-(L1.*cos(A1)));
e=2.*L3.*((L2.*sin(A2))-(L1.*sin(A1)));
f=(L1.^2)+(L2.^2)+(L33.^2)-(L44.^2)-(2.*L1.*L2.*cos(A2-A1));

A33=2.*atan((-e-sqrt((d.^2)+(e.^2)-(f.^2)))/(f-d));

% LOCATION OF POINT E
Ex=Dx+L5.*cos(pi-A5);
Ey=Dy+L5.*sin(pi-A5);
```

```
% THIRD CASE OF 4 BARS
```

```
A5=-A5; A4=-A4;  
[A6,A7]=Caso_4b(L5,L4,L6,L7,A5,A4,'cruzado');
```

```
Hx=Bx+(L33.*cos(A33));  
Hy=By+L33.*sin(A33);
```

```
Fx=Hx-L6.*cos(A6);  
Fy=Hy-L6.*sin(A6);
```

```
Px=Hx+L66.*cos(A6+(pi/2));  
Py=Hy+L66.*sin(A6+(pi/2));
```

```
%% *****VELOCITY ANALYSIS*****  
A4=-A4; A5=-A5;
```

```
W2=2;
```

```
W3=-(L2.*W2.*sin(A2-A4))./(L3.*sin(A3-A4));  
W4=(L2.*W2.*sin(A2-A3))./(L4.*sin(A4-A3));  
W5=W4;
```

```
W33=-(L2.*W2.*sin(A2-A44))./(L33.*sin(A33-A44));  
W44=(L2.*W2.*sin(A2-A33))./(L44.*sin(A44-A33));
```

```
A4=-A4; A5=-A5;  
W6=(L5.*W5.*sin(A5-A7)-L44.*W44.*sin(A44-A7))./(L6.*sin(A6-A7));  
W7=(L44.*W44.*sin(A44)-L5.*W5.*sin(A5)+L6.*W6.*sin(A6))./(L7.*sin(A7));
```

```
%% *****ACCELERATION ANALYSIS*****
```

```
A4=-A4; A5=-A5;
```

```
Alf2=0;  
Alf3=((L2.*(W2.^2).*cos(A2-A4))+L3.*(W3.^2).*cos(A3-A4))-(L4.*(W4.^2))-(L2.*sin(A4-A2))./(L3.*sin(A4-A3));  
Alf4=((L2.*Alf2.*sin(A3-A2))-(L2.*(W2.^2).*cos(A2-A3))-(L3.*(W3.^2))+L4.*(W4.^2).*cos(A4-A3))./(L4.*sin(A3-A4));  
Alf5=Alf4;
```

```
Alf33=((L2.*(W2.^2).*cos(A2-A44))+L33.*(W33.^2).*cos(A33-A44))-(L44.*(W44.^2))-(L2.*sin(A44-A2))./(L33.*sin(A44-A33));  
Alf44=((L2.*Alf2.*sin(A33-A2))-(L2.*(W2.^2).*cos(A2-A33))-(L33.*(W33.^2))+L44.*(W44.^2).*cos(A44-A33))./(L44.*sin(A33-A44));
```

```
A4=-A4; A5=-A5;  
Alf7=((L44.*Alf44.*sin(A44-A6))+L44.*(W44.^2).*cos(A44-A6))+L6.*(W6.^2)-(L5.*Alf5.*sin(A5-A6))-  
(L5.*(W5.^2).*cos(A5-A6))-(L7.*(W7.^2).*cos(A7-A6))./(L7.*sin(A7-A6));  
Alf6=((L5.*Alf5.*sin(A5-A7))+L5.*(W5.^2).*cos(A5-A7))+L7.*(W7.^2)-(L44.*Alf44.*sin(A44-A7))-  
(L44.*(W44.^2).*cos(A44-A7))-(L6.*(W6.^2).*cos(A6-A7))./(L6.*sin(A6-A7));
```

```
A4=-A4; A5=-A5;
```

```
%% *****SIMULATION*****
```

```
for i=1:length(A2)
```

```
plot(Bx,By,'-b');  
hold on
```

```

plot(0,0,'or');
axis([-7 15 -12 6]);
plot(Bx(i),By(i),'or');
plot(Cx(i),Cy(i),'or');
plot(Dx,Dy,'or');
plot(Ex(i),Ey(i),'or');
plot(Hx(i),Hy(i),'or');
plot(Fx(i),Fy(i),'or');
plot(Px(i),Py(i),'or');

```

```

plot([0 Dx],[0 0],'-k','LineWidth',0.001);
plot([0 Bx(i)],[0 By(i)],'-k','LineWidth',1);
plot([Bx(i) Cx(i)],[By(i) Cy(i)],'-k','LineWidth',1);
plot([Dx Cx(i)],[Dy Cy(i)],'-k','LineWidth',1);
plot([Dx Ex(i)],[Dy Ey(i)],'-k','LineWidth',1);
plot([Cx(i) Ex(i)],[Cy(i) Ey(i)],'-k','LineWidth',1);
plot([Bx(i) Hx(i)],[By(i) Hy(i)],'-k','LineWidth',1);
plot([Dx Hx(i)],[Dy Hy(i)],'-k','LineWidth',1);
plot([Ex(i) Fx(i)],[Ey(i) Fy(i)],'-k','LineWidth',1);
plot([Hx(i) Fx(i)],[Hy(i) Fy(i)],'-k','LineWidth',1);

```

```

patch([Dx,Cx(i),Ex(i)],[Dy,Cy(i),Ey(i)],'m')
patch([Hx(i),Px(i),Fx(i)],[Hy(i),Py(i),Fy(i)],'m')

```

```

title('THEO JANSEN MECHANISM');

```

```

set(gcf,'color','W')
set(gca,'xcolor','W')
set(gca,'ycolor','W')
set(gca,'color','W')

```

```

plot(Px(1:i),Py(1:i),'-b');

```

```

[Sopx,Sopy]=Soporte(0,0,5);
plot(Sopx,Sopy,'r')
[Sopx,Sopy]=Soporte(Dx,Dy,5);
plot(Sopx,Sopy,'r')

```

```

text(-1,-0.8,'A');
text(Bx(i)+0.1*L2,By(i)+0.1*L2,'B');
text(Cx(i)+0.1*(L3/2),Cy(i)+0.1*(L3/2),'C');
text(5.8,-0.8,'D');
text(Ex(i)+0.1*L5,Ey(i)+0.1*L5,'E');
text(Hx(i)+0.1*(L3/2),Hy(i)+0.1*(L3/2),'H');
text(Fx(i)+0.1*(L3/2),Fy(i)+0.1*(L3/2),'F');

```

```

pause(0.1);

```

```

hold off;
end

```

```
%% *****GRAPHS FOR POSITION, VELOCITY AND ACCELERATION*****
```

```
figure(2);
subplot(2,2,1),plot(A2,A3,'k',A2,W3,'b',A2,Alf3,'r');
grid on;title('bar L3')

legend('Pos','Vel','Ace');
subplot(2,2,2),plot(A2,A4,'k',A2,W4,'b',A2,Alf4,'r');
grid on;title('bar L4')
legend('Pos','Vel','Ace');
subplot(2,2,3),plot(A2,A33,'k',A2,W33,'b',A2,Alf33,'r');
grid on;title('bar L33')
legend('Pos','Vel','Ace');
subplot(2,2,4),plot(A2,A44,'k',A2,W44,'b',A2,Alf44,'r');
grid on;title('bar L44')
legend('Pos','Vel','Ace');
```

```
figure(3);
set(gcf,'color','W')
set(gca,'xcolor','W')
set(gca,'ycolor','W')
set(gca,'color','W')

subplot(2,2,1),plot(A2,A6,'k',A2,W6,'b',A2,Alf6,'r');
grid on;title('bar L6')
legend('Pos','Vel','Ace');
subplot(2,2,2),plot(A2,A7,'k',A2,W7,'b',A2,Alf7,'r');
grid on;title('bar L7')
legend('Pos','Vel','Ace');
```

```
figure(4);

subplot(2,2,1),plot(Hx,Hy,'k')
grid on;title('Pos H')
xlabel('Hx');ylabel('Hy')
subplot(2,2,2),plot(Px,Py,'b')
grid on;title('Pos P')
xlabel('Px');ylabel('Py')
```

```
figure(5)
subplot(2,2,1),plot(Bx,By,'k')
grid on;title('Pos B')
xlabel('Bx');ylabel('By')
subplot(2,2,2),plot(Cx,Cy,'b')
grid on;title('Pos C')
xlabel('Cx');ylabel('Cy')
subplot(2,2,3),plot(Ex,Ey,'r')
grid on;title('Pos E')
xlabel('Ex');ylabel('Ey')
subplot(2,2,4),plot(Fx,Fy,'g')
grid on;title('Pos F')
xlabel('Fx');ylabel('Fy')
```

```
figure(6);

subplot(2,2,1),plot(A2,Px,'r')
grid on;title('Angle vs Locus')
xlabel('A2');ylabel('Px')
subplot(2,2,2),plot(A2,Py,'g')
grid on;title('Angle vs Locus')
xlabel('A2');ylabel('Py')
```

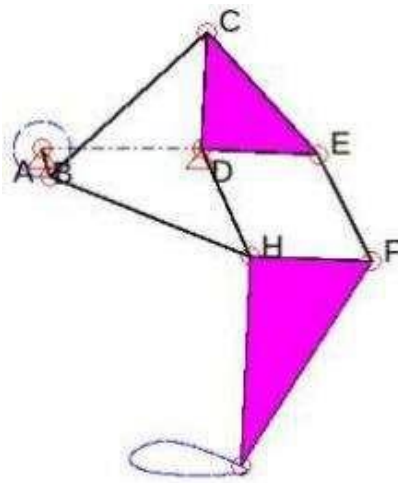
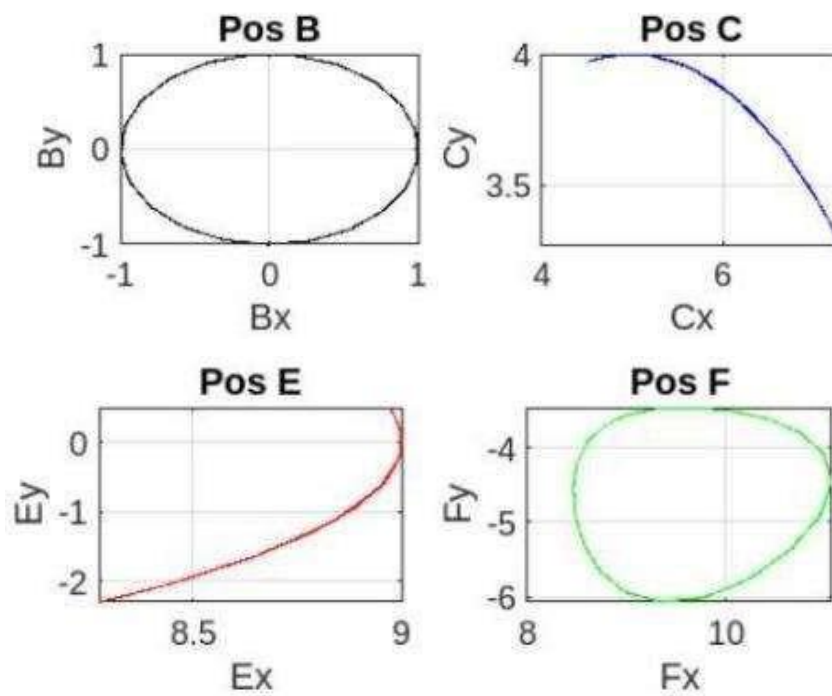


Figure 9. Locus of Leg tip

LOCUS OF POINTS H AND P



LOCUS OF POINTS B,C,E AND F

4.3.2 Motion Analysis

Motion analysis : Motion analysis determines joint reactions and inertial forces acting on the links of the mechanisms in the specified range of motion. All links are treated as rigid bodies. Motion analysis provides complete quantitative information on kinematics (position, velocity, acceleration) and dynamics (forces, moments) of the moving mechanism. Motion analysis in solidworks : Solidworks is capable to accurately simulate and analyze the motion of an assembly while incorporating the effects of Motion Study elements (including forces, springs, dampers, and friction). A Motion Analysis study also calculates loads that can be used to define load cases for structural analyses. A Motion Analysis study combines motion study elements with mates in motion calculations. Consequently motion constraints, material properties, mass, and component contact are included in the SOLIDWORKS Motion kinematic solver calculations. To use the SOLIDWORKS Motion solver in a motion study, select Motion Analysis from the motion studies type list in the MotionManager performing static analysis in solidworks. To perform the static analysis for a given model, the below given steps were followed:-

Create a static study: To access the Study PropertyManager, right-click Simulation study tab and select Create New Simulation Study. Define the properties of the study. Click the Adaptive tab to activate the h- or p-adaptive method to improve results automatically. Define material for each solid, shell, and beam. To define a material for a solid, shell, or beam, right-click its icon in the Simulation study tree and select Apply/Edit Material

Define restraints: Right-click the Fixtures icon in the Simulation study tree and select from the list. Apply adequate restraints to stabilize each body, shell, and component on its own or through contact conditions and connectors linking it with other objects or to the ground. Define external loads. For assemblies and multibody parts, use component contact and contact sets to simulate the behavior of the model.

Mesh the model and run the study. If you run a study before meshing it, the program meshes the study automatically before running it. You can also request to run the study by checking Run analysis after meshing in the Mesh PropertyManager.

4.3.3 Terrain Interaction Analysis

To study the performance of the spider bot on various terrains, simplified contact models of hard, rough and deformable surfaces were created in real. The assembled robot model was positioned on each virtual terrain, and Motion Analysis was performed considering appropriate contact, friction coefficients and gravitational loading. On hard surfaces, a rigid ground plane with high stiffness and moderate friction was defined. For rough terrain, an irregular profile with varying elevations was modeled to replicate stones and uneven ground. For deformable terrain, a lower stiffness support and reduced friction were used to approximate the behavior of mud or loose soil. Time-dependent simulations were run to observe leg trajectories, support polygon evolution and stability margins. The results, including joint reactions, ground reaction

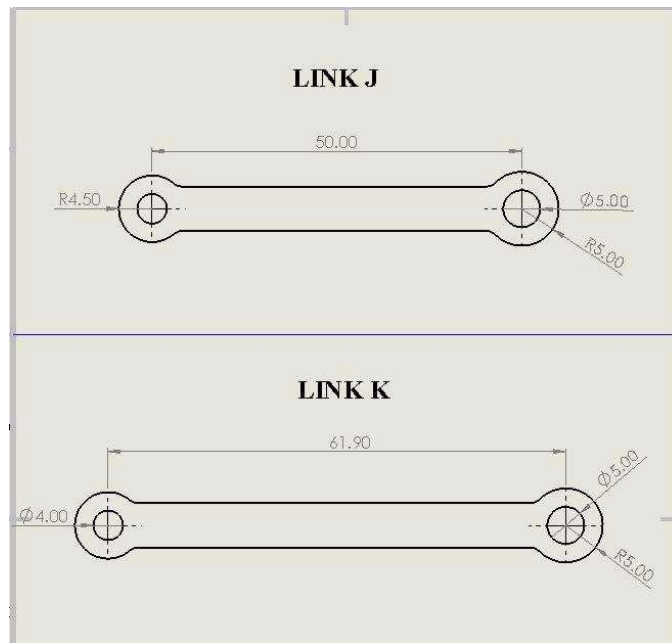
forces and body displacements, were post-processed to correlate with experimental tests carried out on physical terrains, as shown in Figures 18–20.

4.4 Fabrication

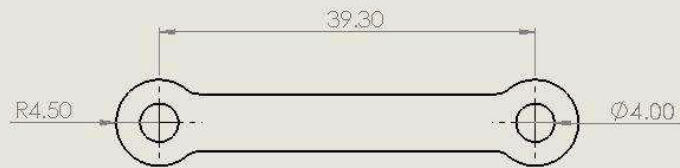
4.4.1 Laser Cutting

Laser cutting is a technology that uses a laser to vaporize materials, resulting in a cut edge. While typically used for industrial manufacturing applications, it is now used by schools, small businesses, architecture, and hobbyists. Laser cutting works by directing the output of a high-power laser most commonly through optics. The laser optics and CNC (computer numerical control) are used to direct the laser beam to the material. A commercial laser for cutting materials uses a motion control system to follow a CNC or G-code of the pattern to be cut onto the material. The focused laser beam is directed at the material, which then either melts, burns, vaporizes away, or is blown away by a jet of gas, leaving an edge with a high-quality surface finish.

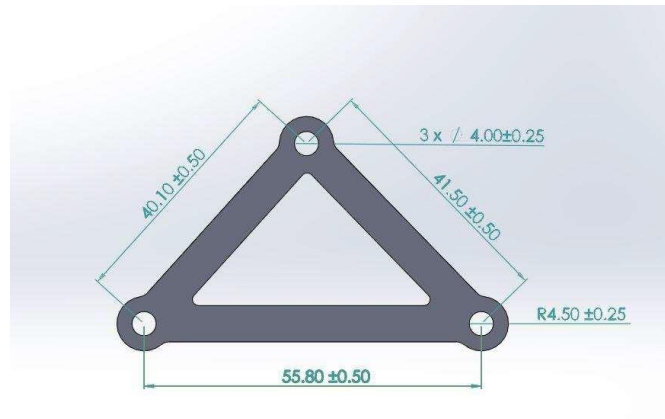
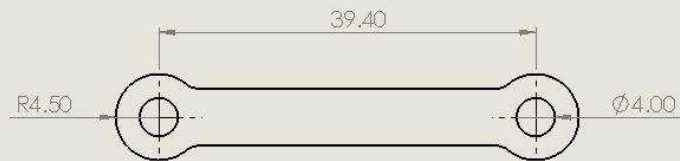
Part dimensions



LINK C



LINK F



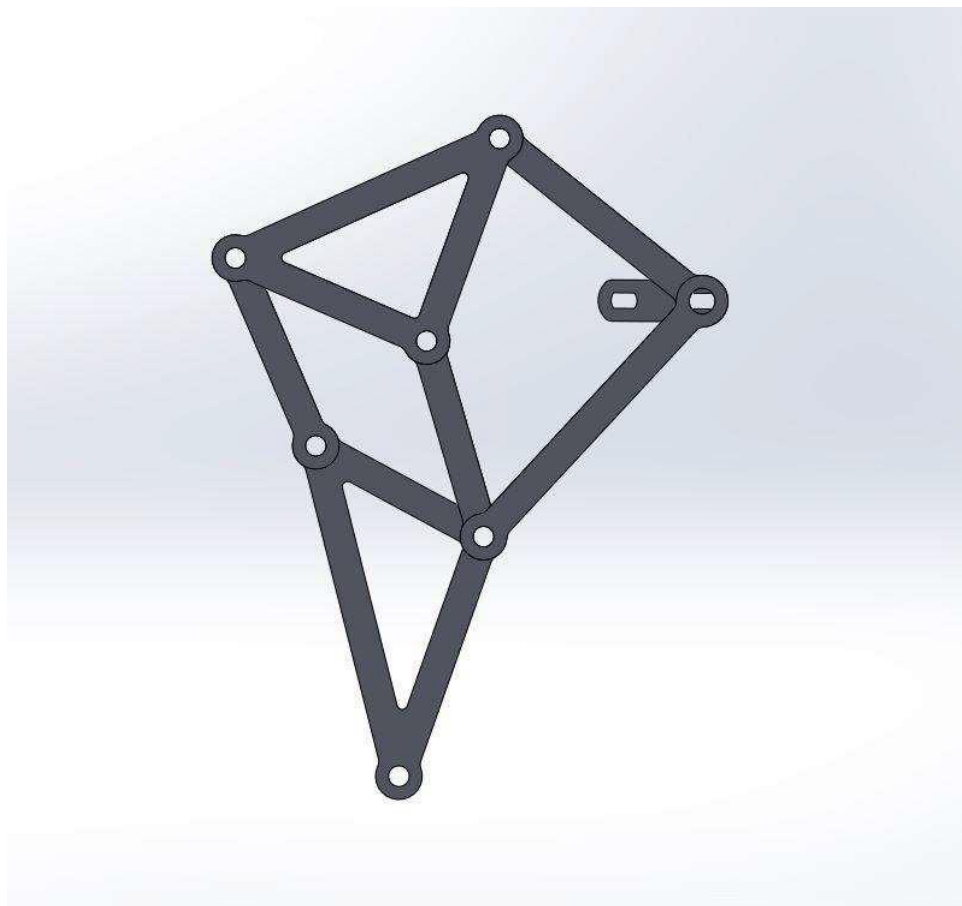
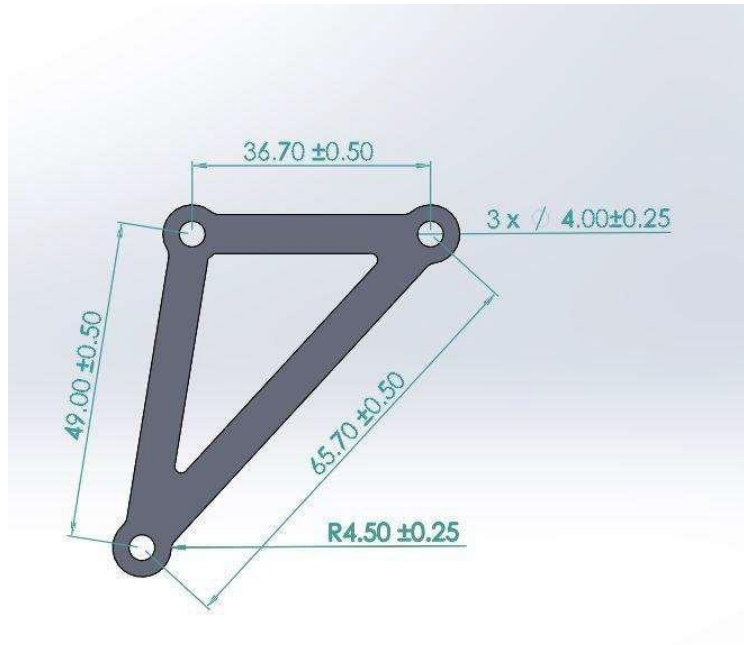


Figure 10. Link Dimensions

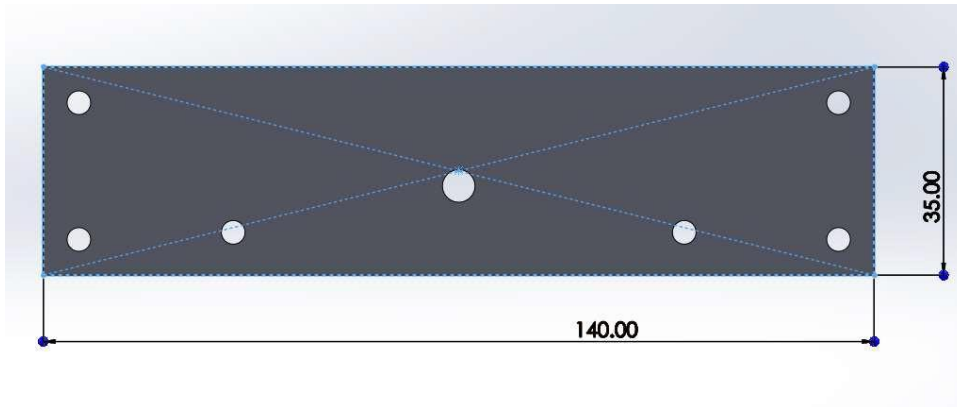


Figure 11. Body Panel Dimensions

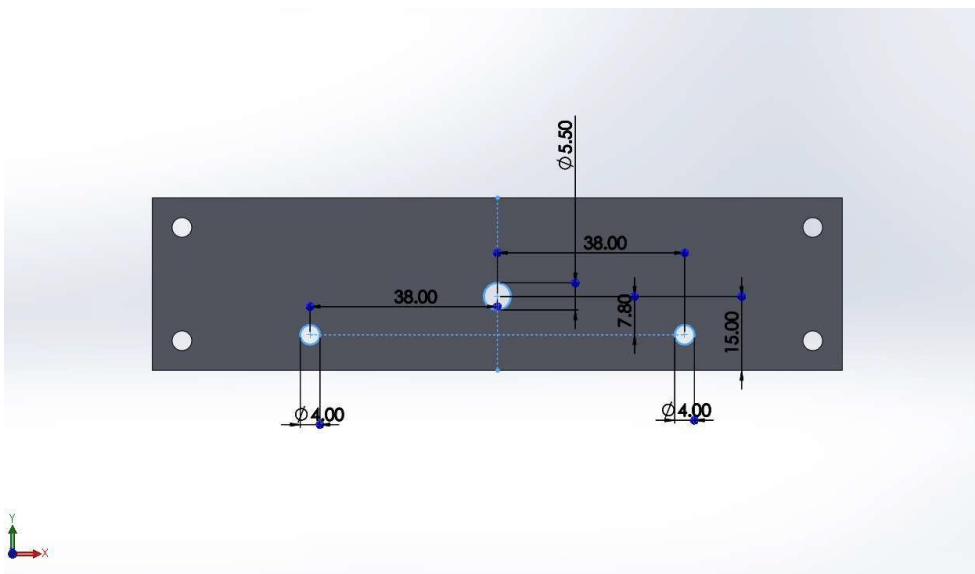


Figure 12. Body Panel Orifice Dimensions



Figure 13. Laser Cut Leg Links



Figure 14. Laser Cut Body Panels

4.4.2 3D Printing

3D printing or additive manufacturing is the construction of a three-dimensional object from a CAD model or a digital 3D model. It can be done in a variety of processes in which material is deposited, joined or solidified under computer control. with material being added together (such as plastics, liquids or powder grains being fused), typically layer by layer. Additive manufacturing or 3D printing has rapidly gained importance in the field of engineering due to its many benefits. Some of these benefits include enabling faster prototyping, reducing manufacturing costs, increasing product customization, and improving product quality.

Furthermore, the capabilities of 3D printing have extended beyond traditional manufacturing, with applications in renewable energy systems. 3D printing technology can be used to produce battery energy storage systems, which are essential for sustainable energy generation and distribution.

Another benefit of 3D printing is the technology's ability to produce complex geometries with high precision and accuracy. This is particularly relevant in the field of microwave engineering, where 3D printing can be used to produce components with unique properties that are difficult to achieve using traditional manufacturing methods.

Leg joint dimensions

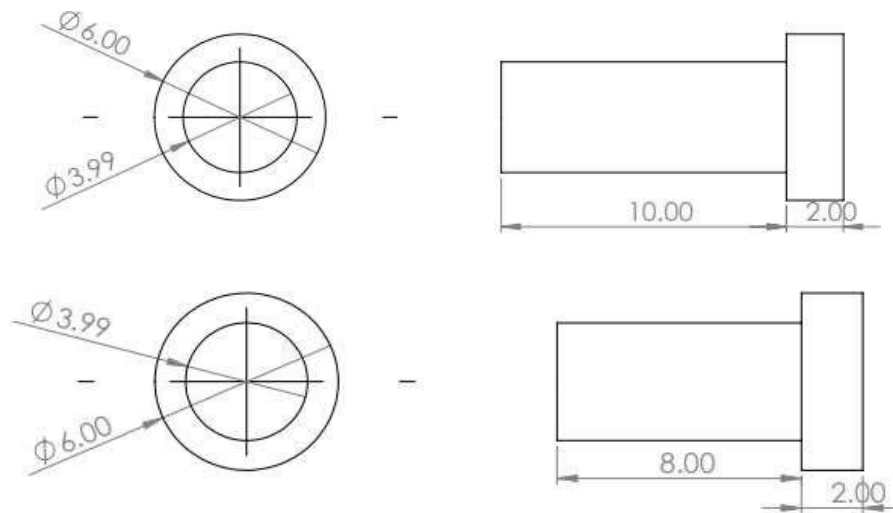


Figure 15. 3D printed Joint dimensions

Shaft dimensions

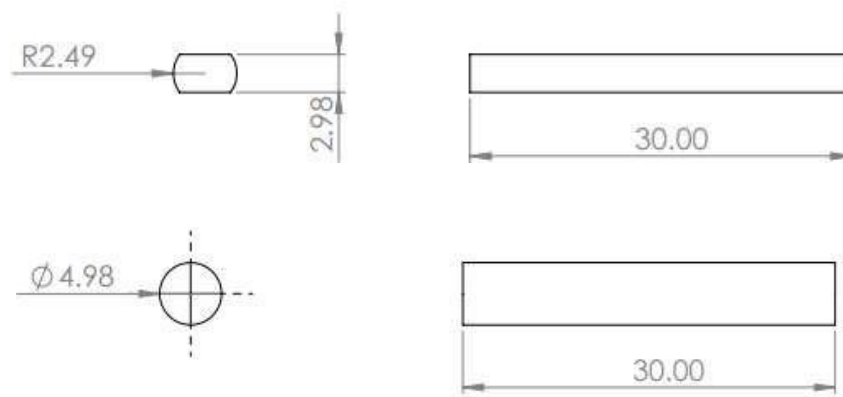


Figure 16. 3D printed shaft dimensions

4.5 Fully Constructed Model

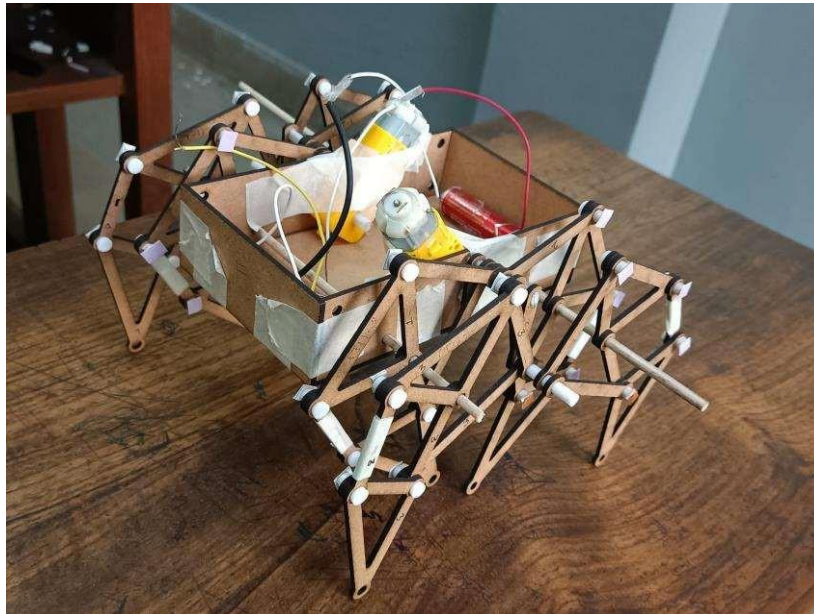


Figure 17. Fully Constructed Model

4.6 Arduino-Based Control and Navigation

To automate the locomotion of the spider bot and to experimentally validate the kinematic and dynamic analysis, an embedded control system was implemented using an Arduino microcontroller. The primary objectives of the control system were to

- (i) provide reliable forward motion at the desired speed,
- (ii) enable in-place turning for navigation over irregular terrain, and
- (iii) ensure stable operation under varying load and surface conditions.

4.6.1 Hardware Integration

The control architecture is built around an Arduino UNO development board, which was selected due to its ease of programming, wide community support and sufficient number of digital I/O pins. Locomotion is achieved using DC geared motors mounted on both sides of the spider bot, each driving a shaft that actuates the Theo-Jansen legs. An L298N H-bridge motor driver module is used to interface the low-power control signals from the Arduino with the higher-current DC motors and the battery supply.

The L298N module provides bidirectional control of two DC motors through four logic inputs (IN1–IN4) and two enable pins (ENA, ENB), which are pulse-width modulated (PWM) to adjust motor speed. A separate battery pack is used to power the motors, while the Arduino board is powered either from USB or a regulated supply. Common ground is maintained between the driver, Arduino and battery pack to ensure proper reference levels for the control signals.

4.6.2 Control Strategy

Differential drive control was employed to achieve forward motion, reverse motion and steering. In this configuration, the left side legs of the robot are driven by one DC motor and the right side legs by another. When both motors rotate in the same direction and at the same speed, the spider bot moves approximately straight ahead. For turning, the relative speeds and directions of the left and right motors are varied. For example, rotating the left motor forward and the right motor backward causes the robot to rotate about its vertical axis, enabling in-place turning.

The basic motion primitives implemented were:

- **Forward:** Both motors rotating forward at equal PWM duty cycle.
- **Backward:** Both motors rotating in reverse at equal PWM duty cycle.
- **Left turn:** Left motor in reverse, right motor forward.
- **Right turn:** Left motor forward, right motor in reverse.
- **Stop:** Both motors disabled (PWM duty cycle set to zero and all H-bridge inputs set low).

These primitives can be combined in sequence to produce more complex trajectories, allowing the spider bot to navigate around obstacles and adapt its path over hard, rough and muddy terrains.

4.6.3 Arduino Program

The navigation logic was implemented in Arduino C/C++ and uploaded to the microcontroller via the Arduino IDE. The code configures the L298N control pins as digital outputs, sets the motor speed using PWM, and executes timed motion sequences for forward motion and turning maneuvers. This code structure also allows easy extension to sensor-based control in future work (e.g., using ultrasonic distance sensors or inertial measurement units), enabling autonomous navigation instead of purely timed motion commands. The program used is shown below:

```
/*
  Spider Bot - Basic Navigation Code
  Hardware: Arduino UNO (or compatible) + L298N + 2 DC motors + battery

  Motion provided:
    - forward(durationMs)
    - backward(durationMs)
    - turnLeft(durationMs)
    - turnRight(durationMs)
    - stopMotors(durationMs)
    - stopMotors() [immediate stop, no delay]

  Assumed connections (change pin numbers if your wiring is different):

  Left motor:
    L298N IN1 -> D8
    L298N IN2 -> D9
    L298N ENA -> D5 (PWM)

  Right motor:
    L298N IN3 -> D10
    L298N IN4 -> D11
    L298N ENB -> D6 (PWM)

  Both motors share common GND with Arduino and battery.
*/

////////////////////
// Pin Definitions //
////////////////////

// Left motor pins
const int IN1 = 8;    // L298N IN1
const int IN2 = 9;    // L298N IN2
```

```

const int ENA = 5;    // L298N ENA (PWM)

// Right motor pins
const int IN3 = 10;   // L298N IN3
const int IN4 = 11;   // L298N IN4
const int ENB = 6;    // L298N ENB (PWM)

// Motor speed (0-255). Adjust based on your motors and battery.
int motorSpeed = 180;

void setup() {
  // Set all control pins as outputs
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(ENA, OUTPUT);

  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(ENB, OUTPUT);

  // Initially stop
  stopMotors();

  // Optional: Serial for debugging
  Serial.begin(9600);
  Serial.println("Spider Bot navigation started.");
}

void loop() {
  // Example demonstration sequence:
  Serial.println("Forward 2 seconds");
  forward(2000);           // move forward for 2 s
  stopMotors(500);        // stop for 0.5 s

  Serial.println("Backward 1.5 seconds");
  backward(1500);         // move backward for 1.5 s
  stopMotors(500);

  Serial.println("Turn left 1 second");
  turnLeft(1000);         // turn left for 1 s
  stopMotors(500);

  Serial.println("Turn right 1 second");
  turnRight(1000);        // turn right for 1 s
  stopMotors(1500);

  // Repeat

```

```

}

////////////////////////////////////
// Motion Functions //
////////////////////////////////////

// Forward for a specified duration (ms)
void forward(unsigned long durationMs) {
    // Left motor forward
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
    analogWrite(ENA, motorSpeed);

    // Right motor forward
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
    analogWrite(ENB, motorSpeed);

    delay(durationMs);
}

// Backward for a specified duration (ms)
void backward(unsigned long durationMs) {
    // Left motor backward
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, HIGH);
    analogWrite(ENA, motorSpeed);

    // Right motor backward
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
    analogWrite(ENB, motorSpeed);

    delay(durationMs);
}

// Turn left (approx. on the spot) for a specified duration (ms)
void turnLeft(unsigned long durationMs) {
    // Left motor backward
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, HIGH);
    analogWrite(ENA, motorSpeed);

    // Right motor forward
    digitalWrite(IN3, HIGH);
    digitalWrite(IN4, LOW);
    analogWrite(ENB, motorSpeed);
}

```

```

    delay(durationMs);
}

// Turn right (approx. on the spot) for a specified duration (ms)
void turnRight(unsigned long durationMs) {
    // Left motor forward
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
    analogWrite(ENA, motorSpeed);

    // Right motor backward
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, HIGH);
    analogWrite(ENB, motorSpeed);

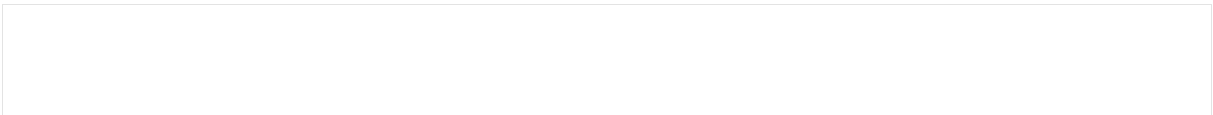
    delay(durationMs);
}

// Stop both motors immediately (no delay)
void stopMotors() {
    // Disable left motor
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);
    analogWrite(ENA, 0);

    // Disable right motor
    digitalWrite(IN3, LOW);
    digitalWrite(IN4, LOW);
    analogWrite(ENB, 0);
}

// Stop both motors and hold that state for durationMs
void stopMotors(unsigned long durationMs) {
    stopMotors();
    delay(durationMs);
}

```



4.7. Testing

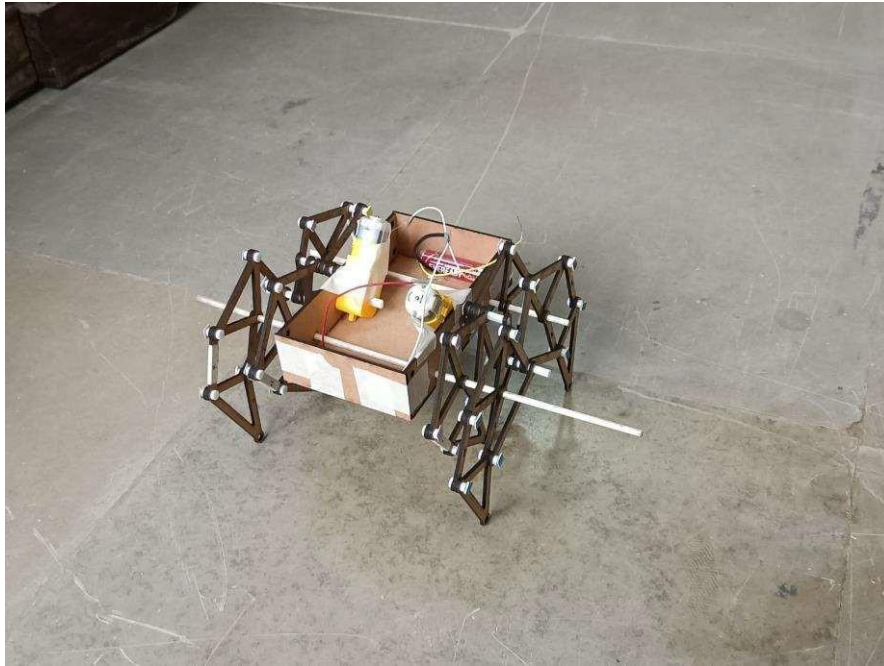


Figure 18. Testing on hard surface



Figure 19. Testing on Rough Terrain



Figure 20. Testing on Muddy terrain

Inference: From Testing we inferred that the spider bot performs in the best on hard surface exhibiting stability without significant slippage followed by Satisfactory performance on rough terrains. Lastly , on deformable surfaces like muddy terrain the lack of reaction forces hindered the movement of the spider bot.

Chapter 5

Results

The objective of this study was to investigate how a walking robot based on the Theo Jansen mechanism performs on various surfaces and to analyze the impact of surface properties on its locomotion characteristics. A series of experiments and simulations were conducted using the designed spider bot. The robot's performance was assessed through SolidWorks motion and structural analysis, MATLAB-based kinematic analysis of the leg mechanism, and physical testing on hard, rough and muddy terrains.

The MATLAB kinematic analysis of the Theo-Jansen leg mechanism provided the locus of the leg tip for one full crank rotation. The resulting trajectory confirmed a clear separation between the support phase (approximately flat, near-horizontal portion of the locus) and the swing phase (curved portion lifted above the ground). This validated that the chosen link dimensions and crank radius generate a nearly straight foot path during stance, which is desirable for stable walking and reduced slip. These results were used to finalize the geometric parameters of the leg and to corroborate the motion obtained from SolidWorks simulations.

In SolidWorks, static and motion analyses were performed on the assembled model. The leg links and body frame were subjected to typical loading conditions corresponding to the robot's self-weight and contact reactions from the ground. Stress and displacement plots showed that the selected material (fibreboard for the frame and 3D-printed plastic for the joints) could safely withstand the operating loads, with maximum stresses remaining below the material yield strengths. Motion simulations on simplified representations of hard, rough and deformable terrain highlighted the changes in body vibrations, leg contact timing and stability margins as surface stiffness and roughness were varied.

The physical prototype was then fabricated using laser-cut fibreboard body panels and Theo-Jansen leg links, along with 3D-printed joints and shafts. An Arduino UNO microcontroller, L298N motor driver, DC motors and on-board batteries were integrated to realize forward motion and turning through differential drive. The Arduino control program successfully executed pre-defined motion sequences, allowing the robot to move forward, reverse and turn on command.

Hard Surfaces:

The robot demonstrated excellent performance on hard floors and rigid surfaces. It maintained good stability, executed precise movements and achieved a high walking speed with negligible slippage. The observed locomotion matched well with the predicted kinematic locus and SolidWorks simulations, confirming the effectiveness of the Theo Jansen leg geometry on firm terrain.

Rough Terrains:

On uneven and rough terrains, the robot exhibited remarkable adaptability. It successfully navigated small obstacles and variations in elevation, with only minor reductions in walking speed. The compliant nature of the leg mechanism and multiple points of support improved ground contact, helping the robot maintain a consistent gait. The experimental observations agreed with the qualitative trends from the simulated rough terrain models.

Deformable Substrates (Muddy Ground):

The robot showed limited success on highly deformable substrates such as muddy ground. The reduced surface stiffness and lower reaction forces led to partial sinking of the legs and increased slippage, which hindered forward progression. Although the robot could still move, its speed and stability were noticeably decreased compared to hard and moderately rough surfaces. These results highlight the importance of surface stiffness and friction in the

performance of Theo Jansen-based walking robots and indicate potential areas for future design improvements, such as wider feet or alternative materials for better traction.

Overall, the combination of MATLAB kinematic analysis, SolidWorks simulations and experimental tests on multiple terrains provided a comprehensive understanding of the spider bot's behavior. The results confirm that the designed Theo Jansen mechanism, together with the implemented Arduino-based control system, is capable of achieving stable and efficient walking on hard and moderately rough surfaces, while also revealing the limitations encountered on highly deformable terrains.

Chapter 6

Conclusion

In this work, a spider bot utilizing the Theo Jansen leg mechanism was designed, analyzed, fabricated and tested on different terrains. The primary objective was to study the locomotion characteristics of a multi-legged walking robot and to evaluate the influence of surface properties on its performance, while using a mechanically efficient linkage-based leg mechanism.

Initially, detailed design calculations were carried out for the Theo Jansen leg mechanism, including link dimensions, joint forces and bending moments in the critical members. The leg links and body frame were then modeled in SolidWorks, where static and motion studies were performed to assess stresses, deformations and kinematic behavior. The results of these simulations confirmed that the selected materials (fibreboard for the frame and 3D-printed plastic for the joints) and dimensions provided sufficient strength and stiffness for the expected loads.

Kinematic analysis of the leg was further refined using MATLAB by generating the locus of the leg tip over one complete cycle. The trajectory clearly showed a nearly straight stance phase close to the ground and a curved swing phase above the ground, validating that the chosen geometry of the Theo Jansen mechanism produced the desired walking pattern. This numerical analysis was in good qualitative agreement with the SolidWorks motion study and the observed motion of the physical prototype.

The spider bot was fabricated using laser-cut leg links and body panels, with 3D-printed joints and shafts to realize the designed mechanism. An Arduino-based control system was implemented, consisting of an Arduino board, L298N motor driver, DC motors and on-board batteries. Using a simple but effective Arduino control program, the robot was able to move forward, reverse and perform in-place turns through differential drive, enabling basic navigation over test terrains.

Experimental testing on hard, rough and muddy surfaces demonstrated that the robot performs best on hard terrain, where it exhibited stable motion, minimal slippage and good agreement with the predicted kinematic and simulation results. On rough terrain, the spider bot maintained acceptable stability and was able to negotiate moderate unevenness, although with slightly increased vibrations and reduced speed. On muddy or highly deformable surfaces, the performance was limited due to reduced support and increased sinking and slippage of the legs, highlighting the sensitivity of Theo Jansen-type walkers to ground stiffness and friction.

Overall, the project successfully demonstrated the feasibility of using the Theo Jansen mechanism for a spider-like walking robot and established a complete workflow from analytical design and CAD-based simulation to fabrication, control and experimental validation. The integration of MATLAB kinematic analysis, SolidWorks simulations and Arduino-based navigation provided a comprehensive understanding of the robot's behavior. Future work could focus on optimizing leg geometry for better performance on deformable terrains, improving traction through modified foot designs or materials, and adding sensors and feedback control to achieve autonomous navigation and adaptive gait control.

Chapter 7

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